



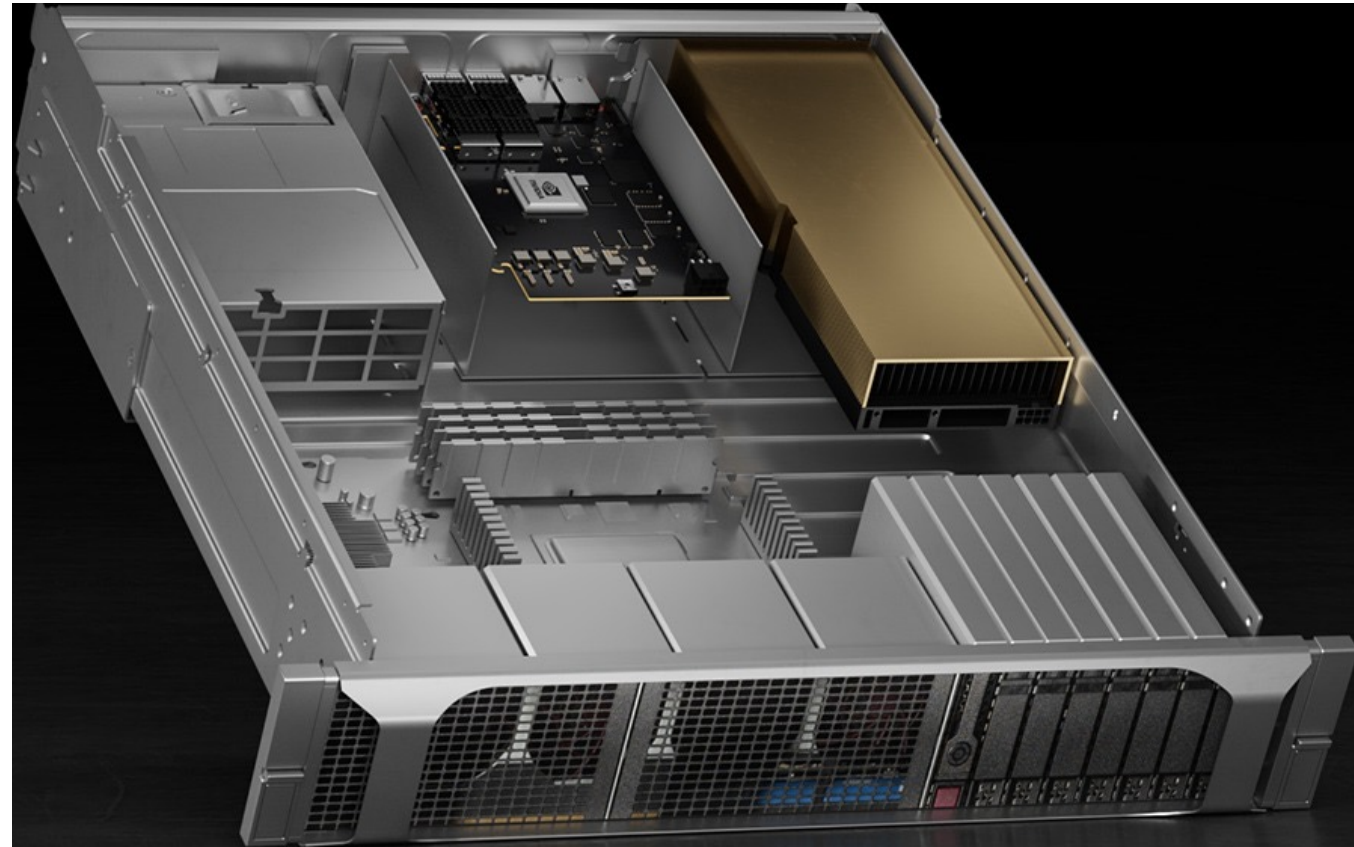
GPUs

2025

André Pereira

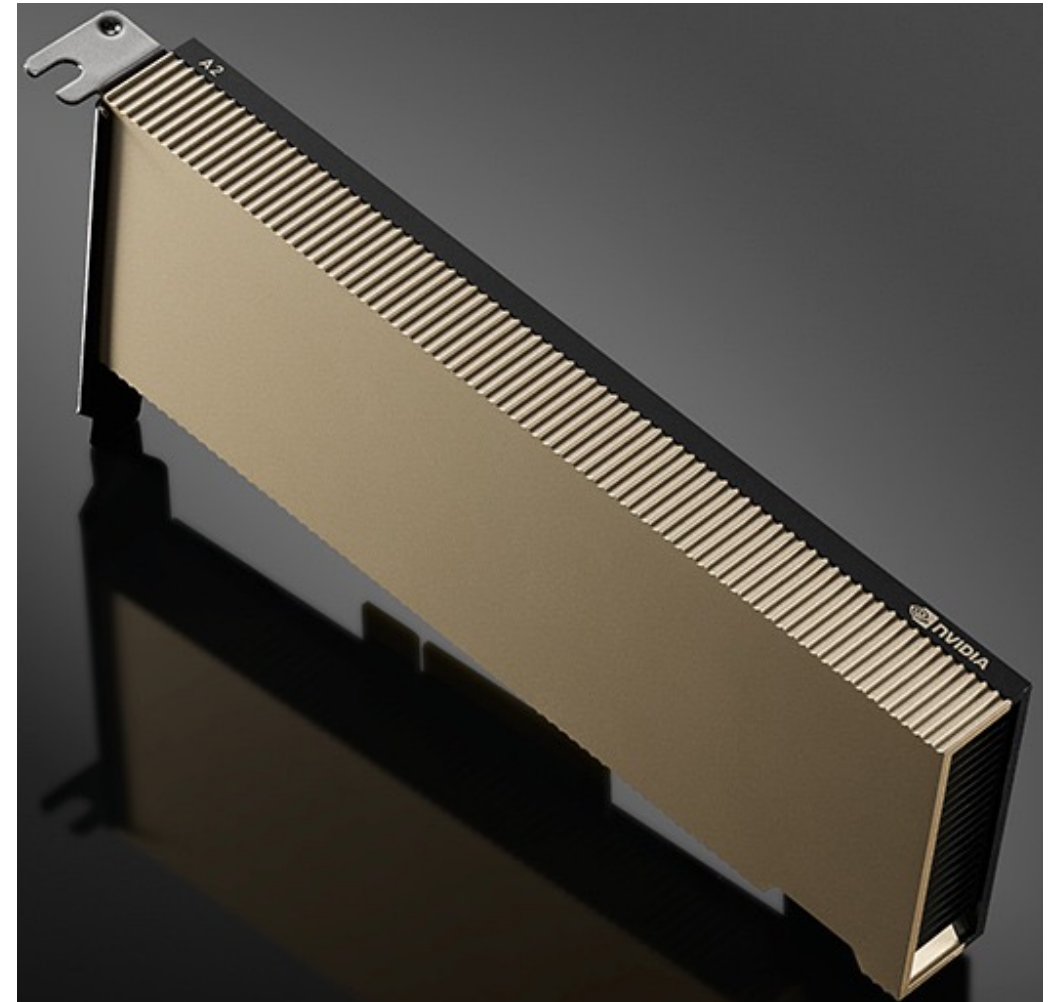


- **A100 40GB and 80GB versions**
 - 6,912 CUDA cores
 - 432 Tensor cores
- **HBM2e**
 - 2 TB/s bandwidth
- **Interconnect**
 - **NVLink:** 3rd generation NVLink (600 GB/s bidirectional)
 - PCIe Gen4





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Deucalion's Nvidia A100 GPU



Peak FP64 ¹	9.7 TFLOPS
Peak FP64 Tensor Core ¹	19.5 TFLOPS
Peak FP32 ¹	19.5 TFLOPS
Peak FP16 ¹	78 TFLOPS
Peak BF16 ¹	39 TFLOPS
Peak TF32 Tensor Core ¹	156 TFLOPS 312 TFLOPS ²
Peak FP16 Tensor Core ¹	312 TFLOPS 624 TFLOPS ²
Peak BF16 Tensor Core ¹	312 TFLOPS 624 TFLOPS ²
Peak INT8 Tensor Core ¹	624 TOPS 1,248 TOPS ²
Peak INT4 Tensor Core ¹	1,248 TOPS 2,496 TOPS ²

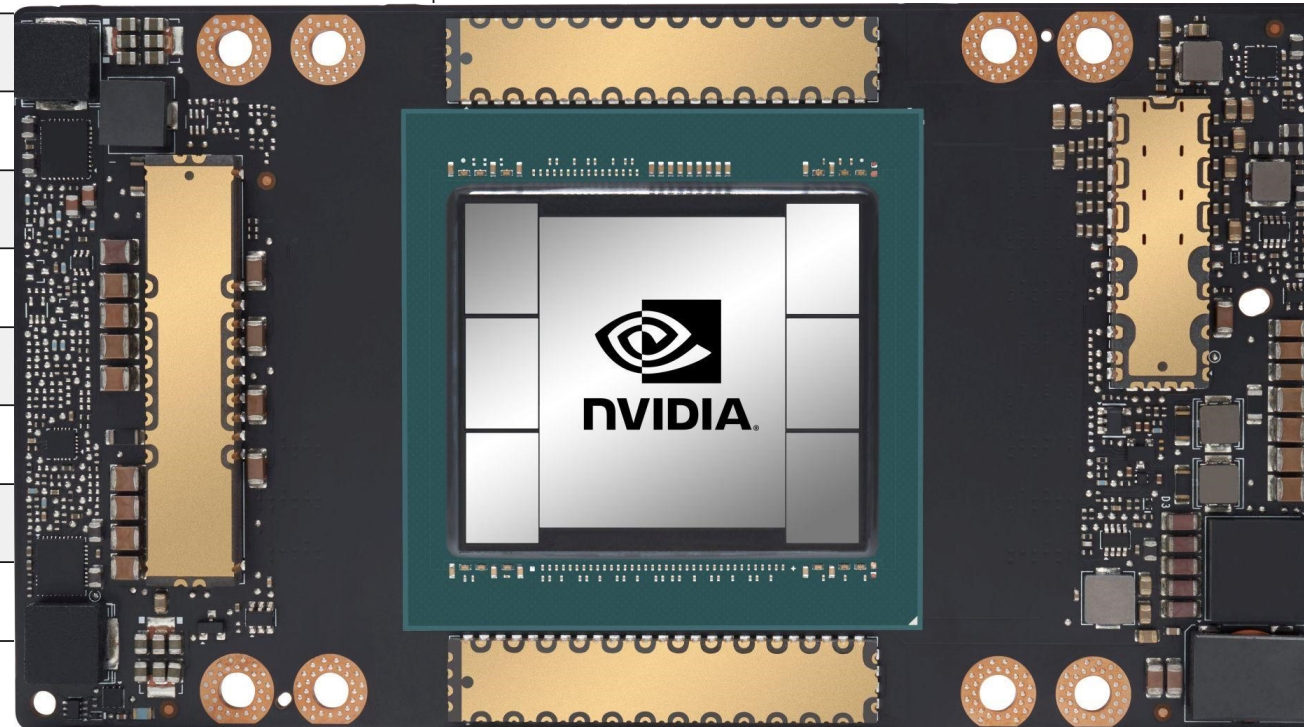
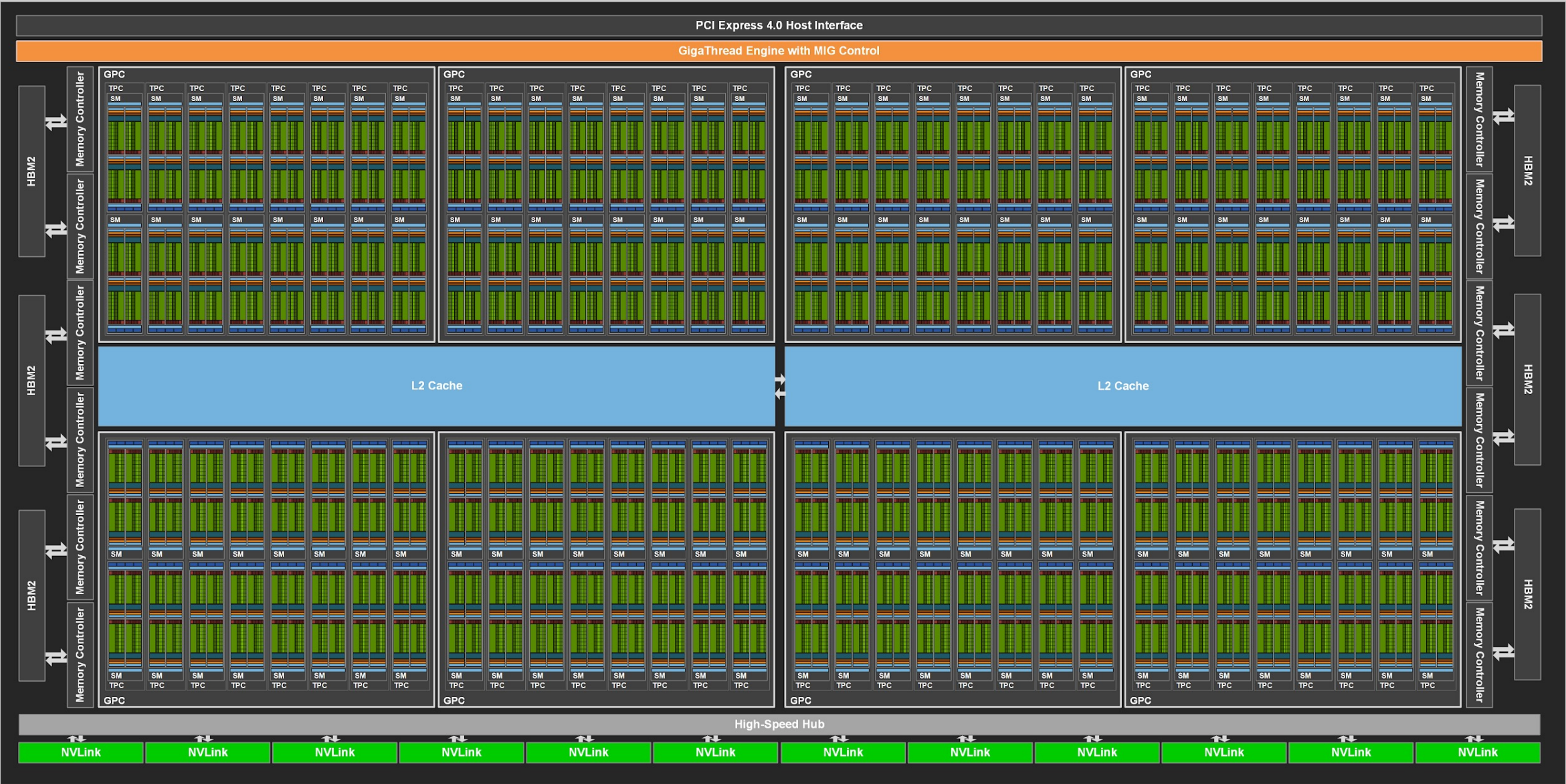


Table 1. A100 Tensor Core GPU performance specs.
1) Peak rates are based on the GPU boost clock.
2) Effective TFLOPS / TOPS using the new Sparsity feature.

A100 Architecture



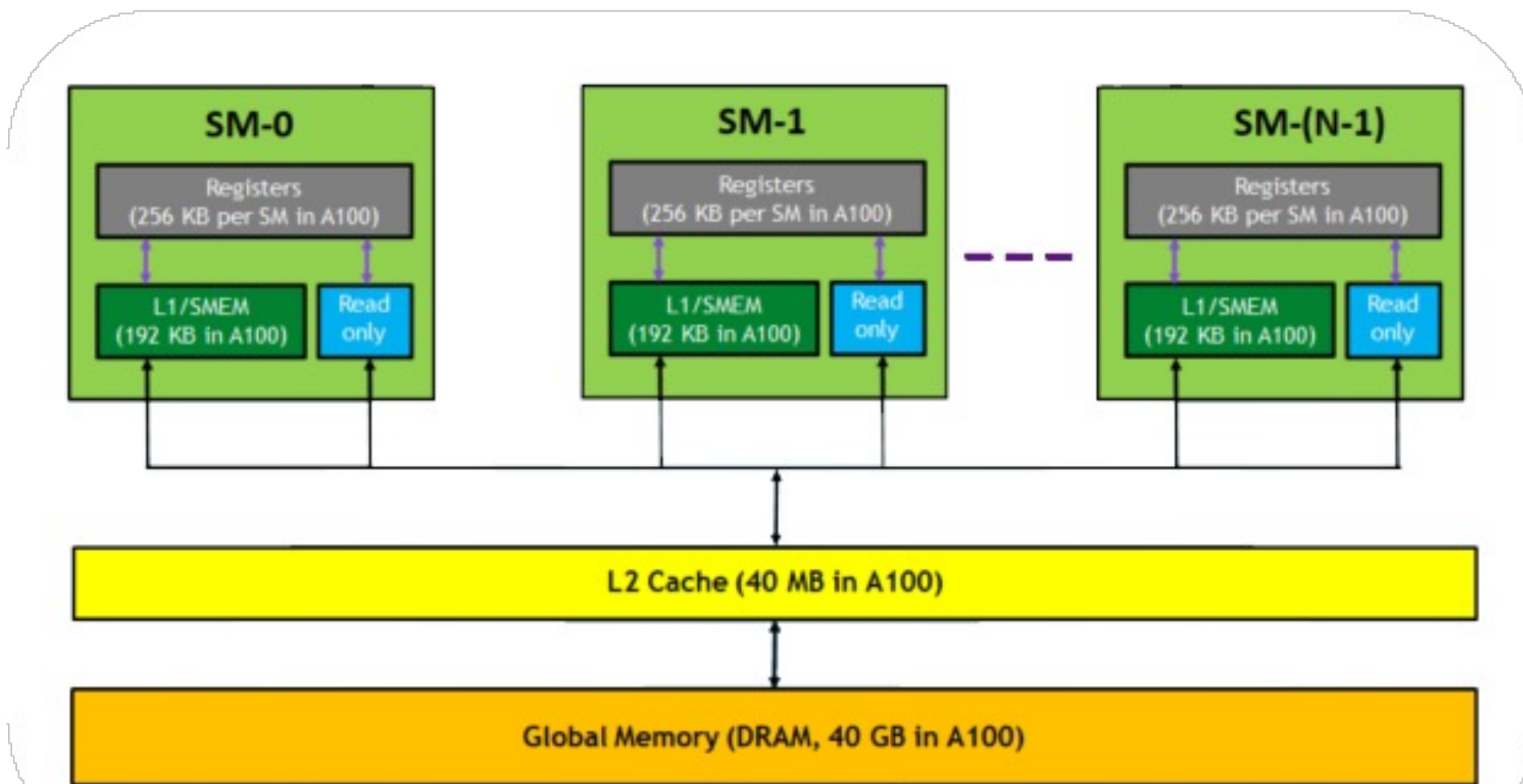
A100 Architecture



- Third-generation Tensor Cores:
 - FP16, BF16, TF32, FP64, INT8, INT4
 - New Tensor Core sparsity feature exploits fine-grained structured sparsity
 - FP16/FP32 and BF16/FP32 mixed-precision Tensor Core operations
- 192 KB of combined shared memory and L1 data cache.



Memory Hierarchy

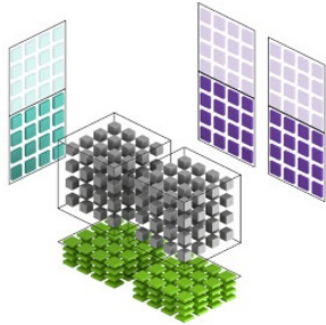




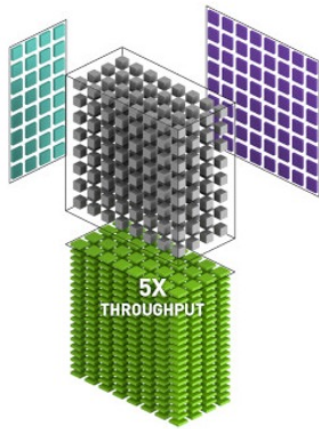
- Asynchronous copy loads data directly from global memory into shared memory
 - Optionally bypassing L1 cache and intermediate register file (RF) usage.
- Instructions for L2 cache management and residency controls
- Warp-level reduction instructions - CUDA Cooperative Groups

Tensor Computing – Does Vectorisation Stand a Chance?

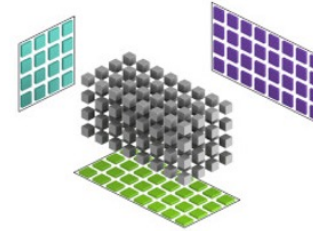
NVIDIA V100 Tensor Core FP16



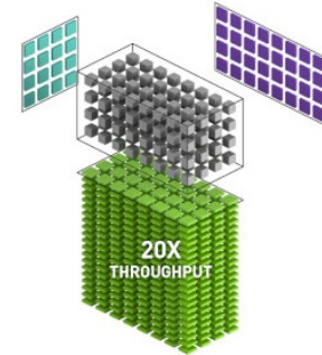
NVIDIA A100 Tensor Core FP16 with Sparsity



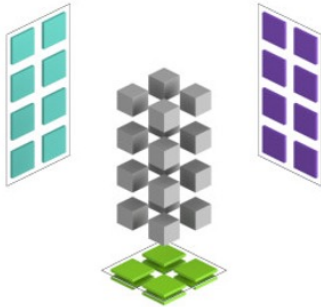
NVIDIA V100 FP32



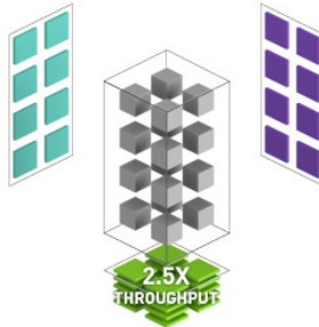
NVIDIA A100 Tensor Core TF32 with Sparsity



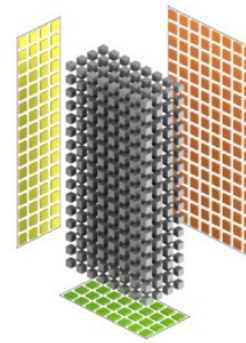
NVIDIA V100 FP64



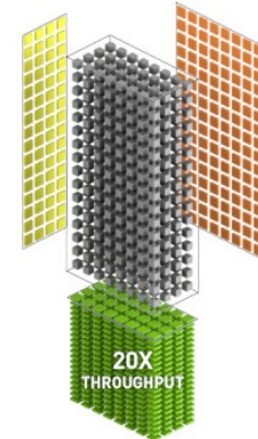
NVIDIA A100 Tensor Core FP64



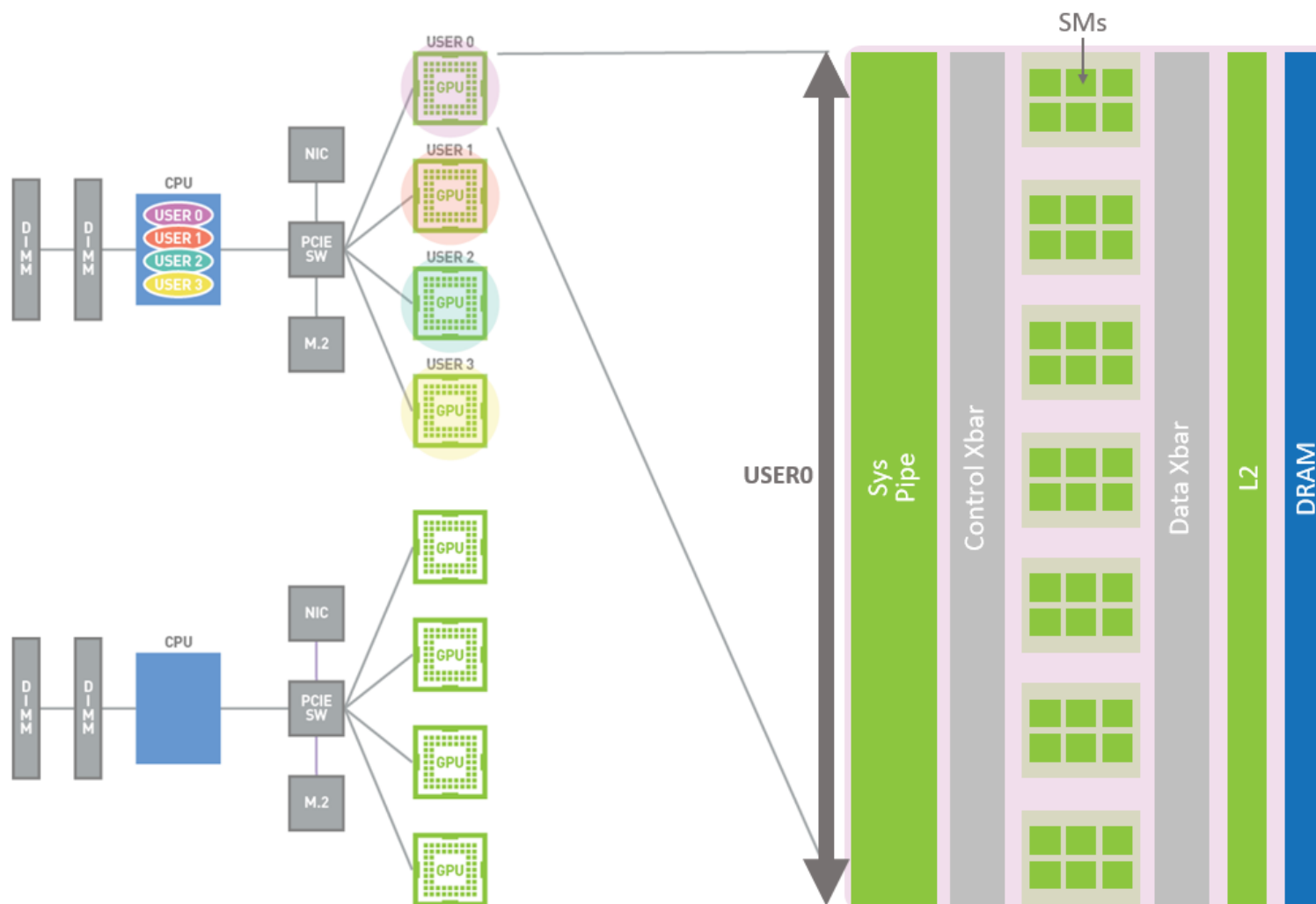
NVIDIA V100 INT8



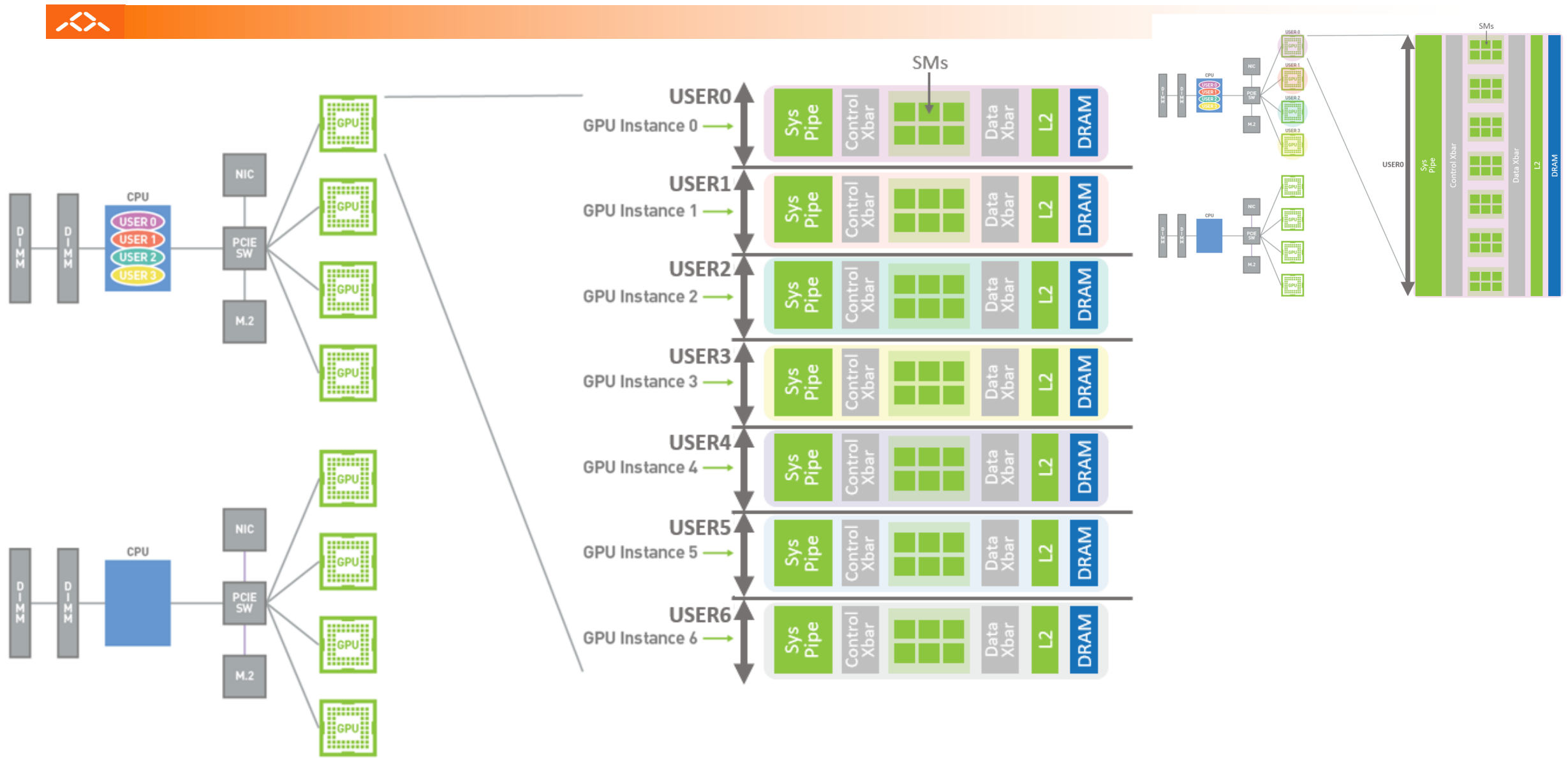
NVIDIA A100 Tensor Core INT8 with Sparsity



Multi Instance GPU



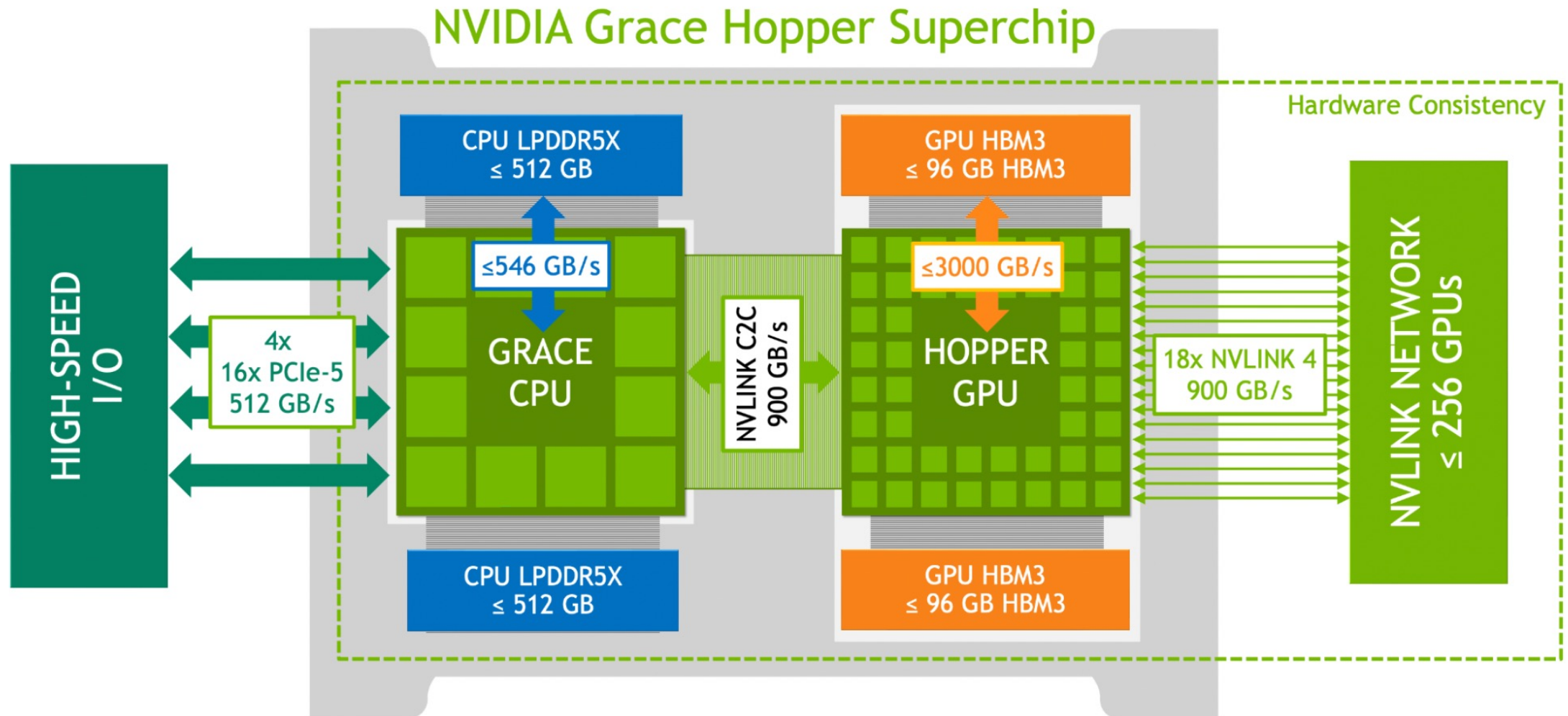
Multi Instance GPU



Grace Hopper Superchip – the Next Gen Heterogeneous Platform



Grace Hopper Superchip



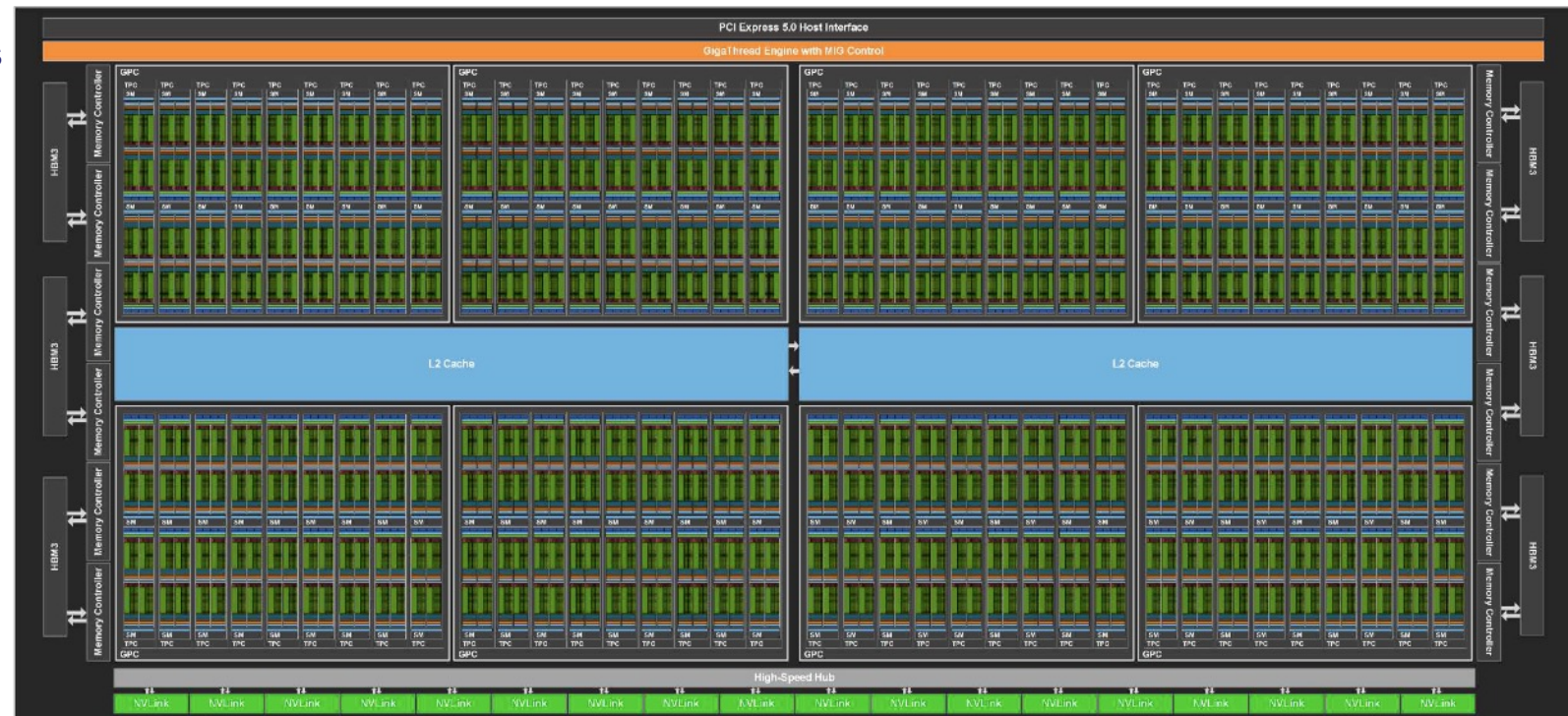
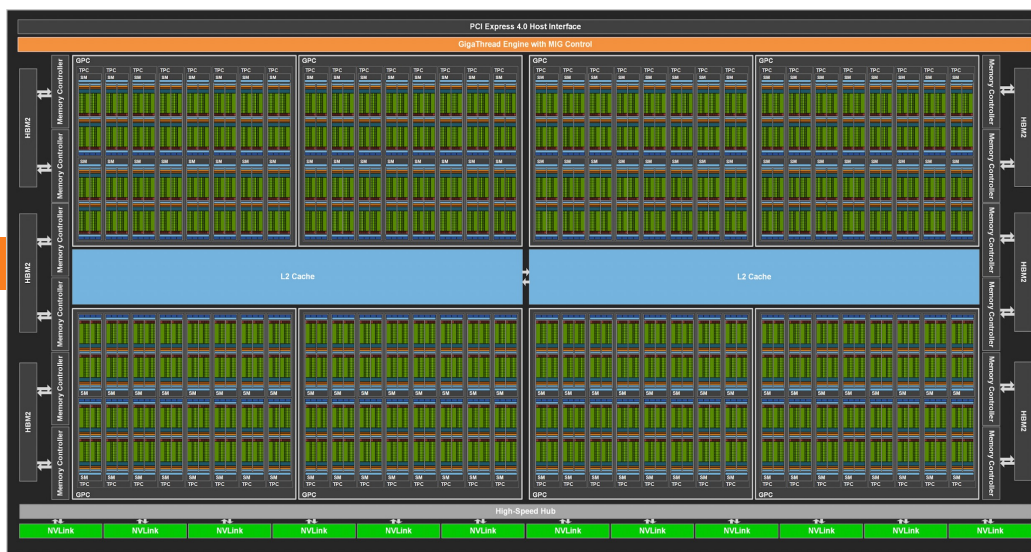


From A100 to Hopper: up to 8 GPC, 132 SMs total

Hopper: NVidia GH100
16 896 FP32 CUDA Cores
528 4th generation Tensor Cores
5 HBM3 (up to 80 GiB)
10 512-bit mem controllers
announced March'22

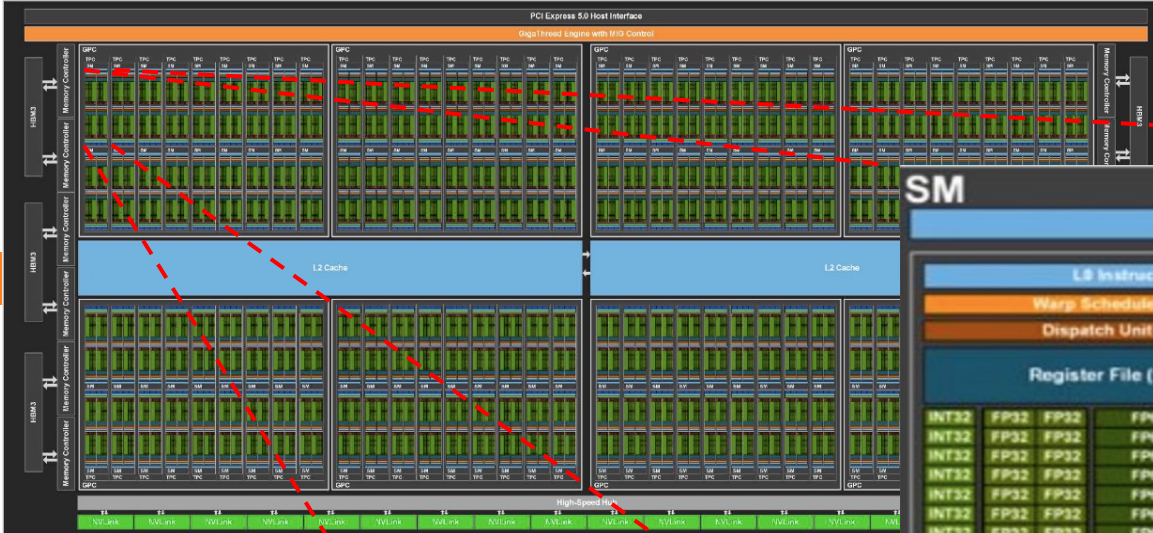
Ampere:
128 SM
8192 CUDA-cores
May'20

Hopper:
GH100
w/ 8 GPC
full w/ 144 SMs





Hopper Architecture



Hopper SM:

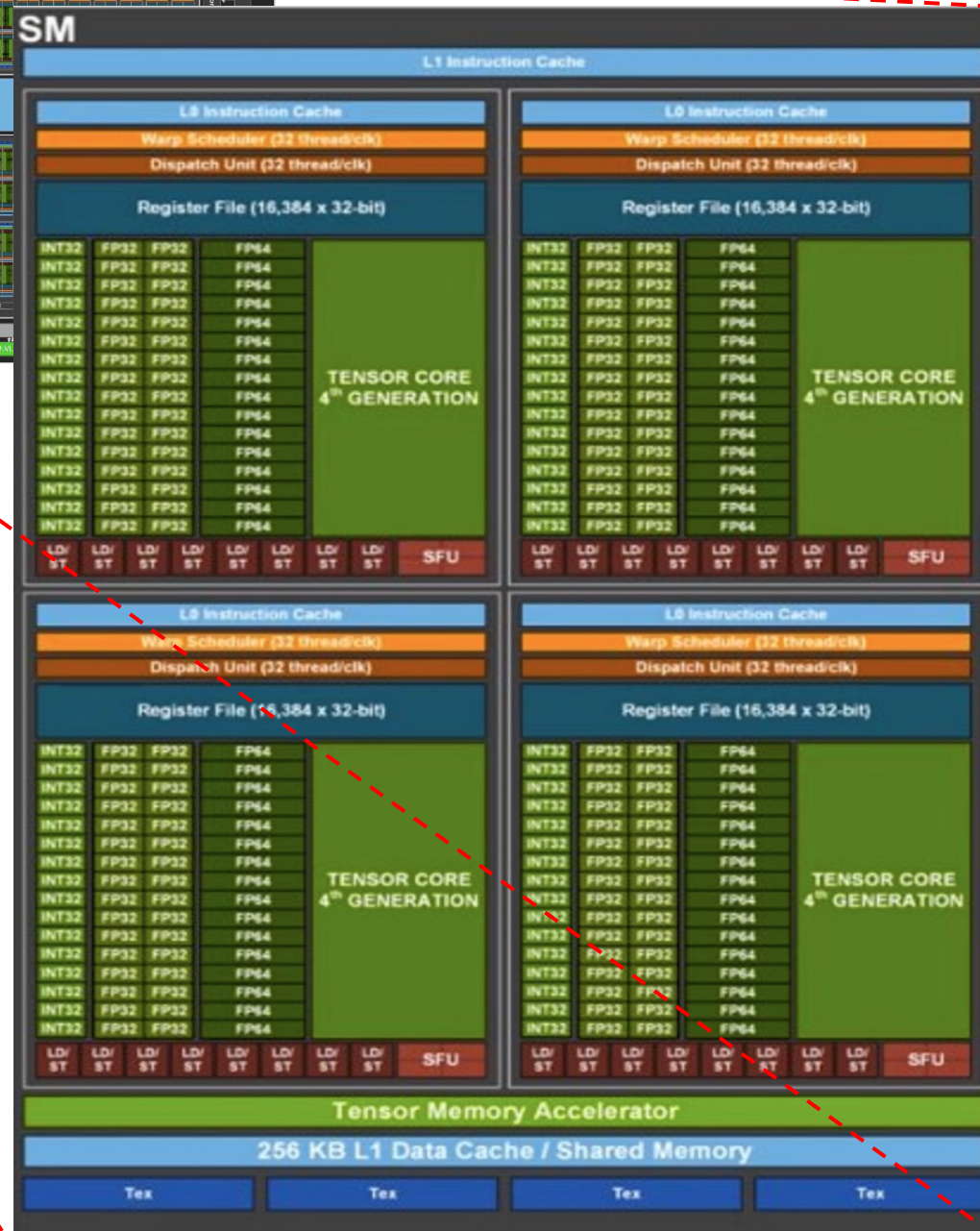
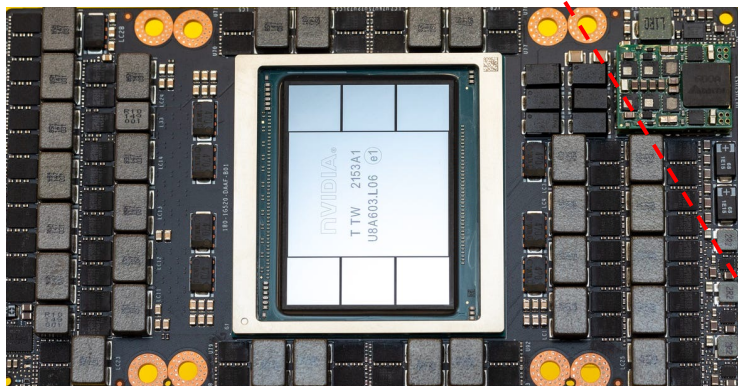
128x FP32 CUDA Cores/SM

64x FP64 CUDA Cores/SM

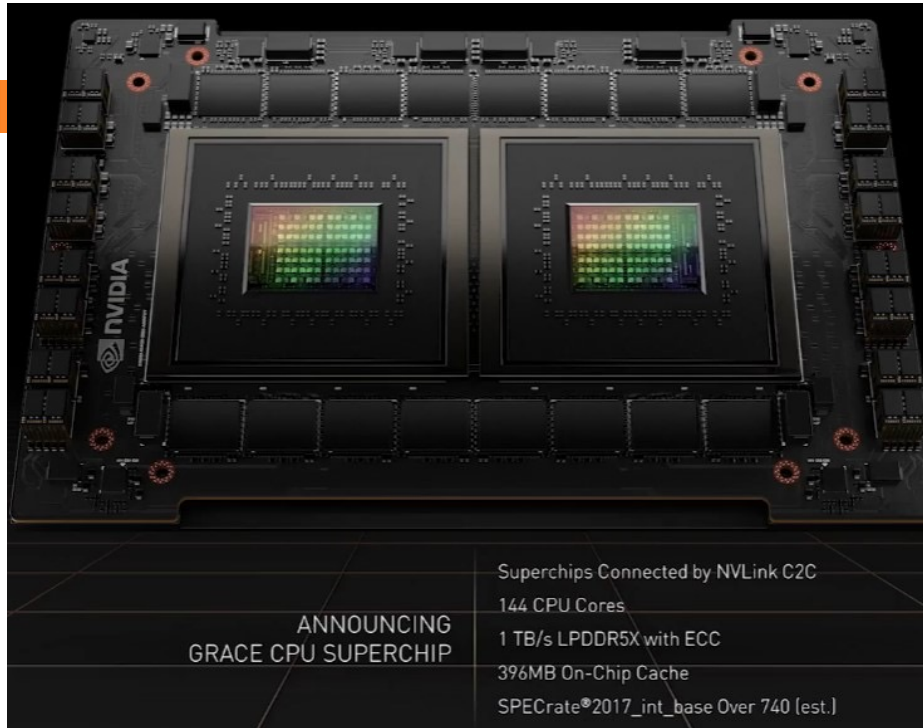
4x 4th generation Tensor Cores

Tensor Cores support (2x faster)
FP64, FP32, TF32, FP16, BF16, **FP8**, INT8...

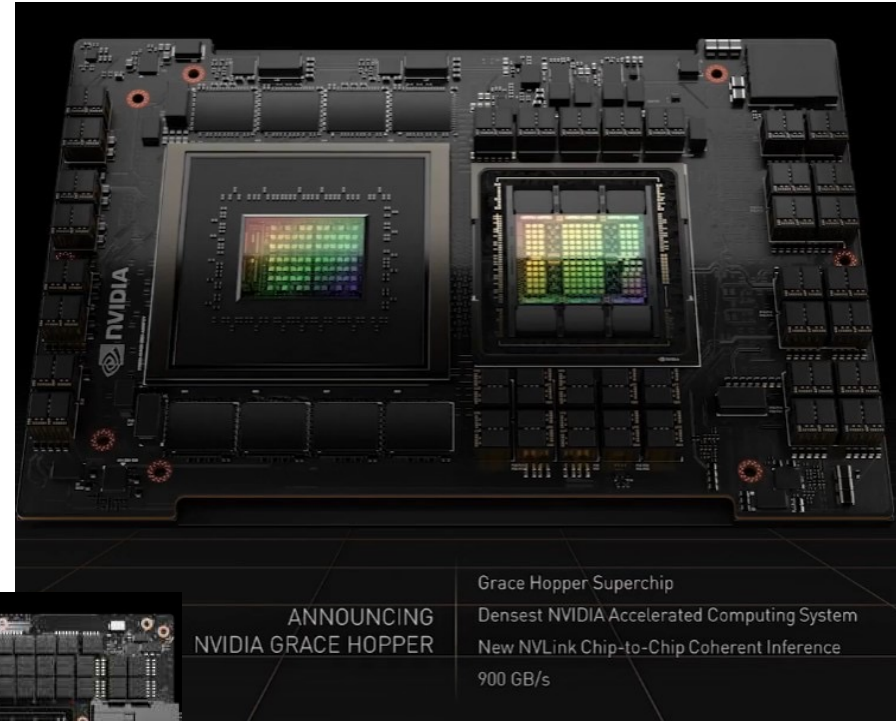
1024 dense FP16/FP32 FMA op's/cycle



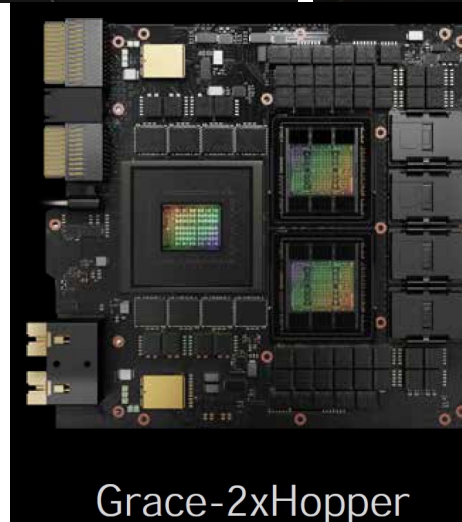
The NVidia Grace & Grace-Hopper Superchips



Grace Superchip:
2x 72-core Arm v9.0
4 nm TSMC
SVE2 support



Grace-Hopper Superchip

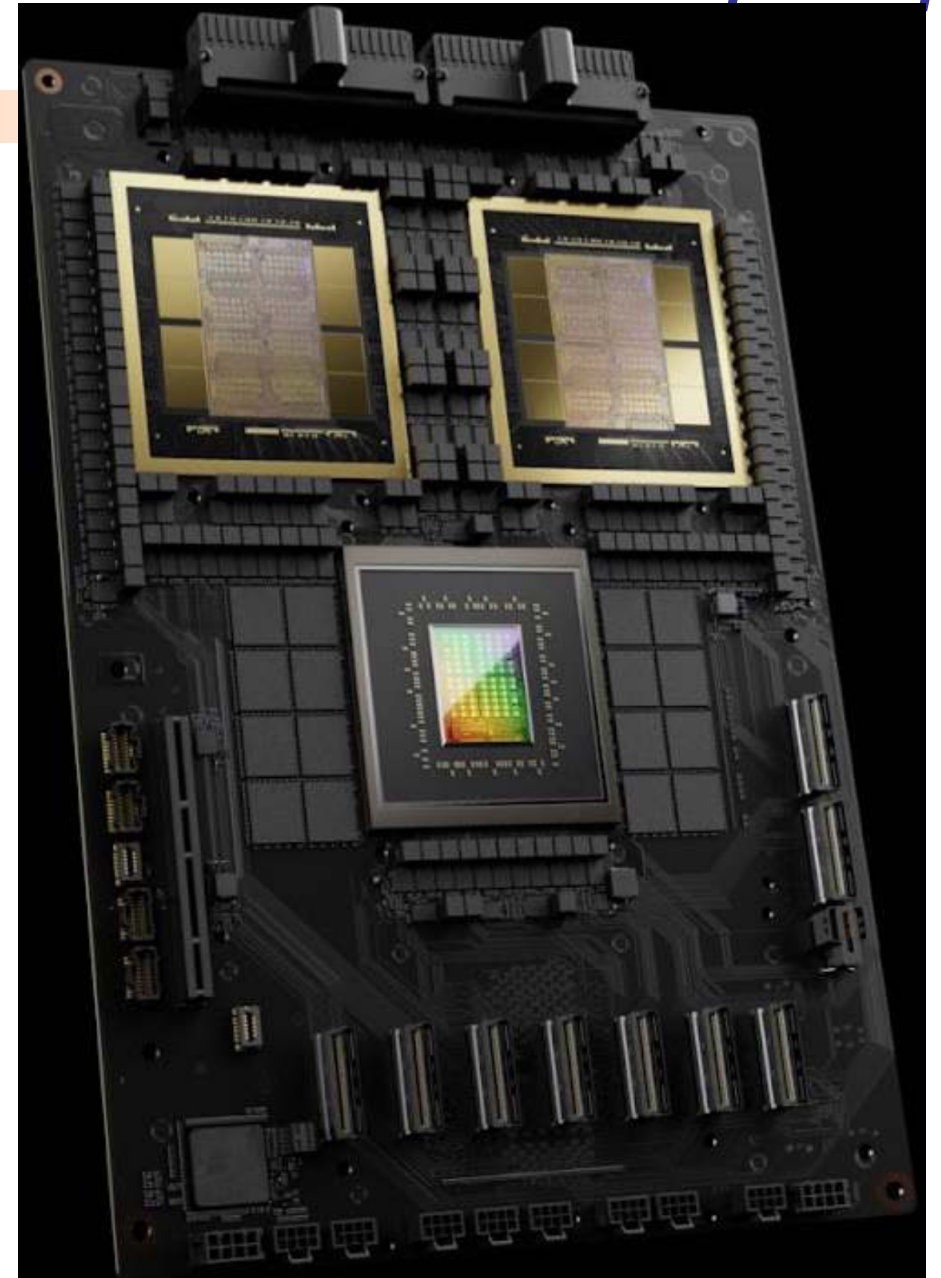


Blackwell Superchip



DGX B200 Technical Specifications

GPU	8x NVIDIA Blackwell GPUs
GPU Memory	1,440 GB total, 64 TB/s HBM3e bandwidth
Performance	FP4 Tensor Core: 144 72* petaFLOPS FP8 Tensor Core: 72 36* petaFLOPS
NVIDIA® NVSwitch™	2x
NVIDIA NVLink Bandwidth	14.4 TB/s aggregate bandwidth
System Power Usage	~14.3 kW max
CPU	2 Intel® Xeon® Platinum 8570 Processors 112 Cores total, 2.1 GHz (Base), 4 GHz (Max Boost)
System Memory	2 TB, configurable to 4 TB
Networking	4x OSFP ports serving 8x single-port NVIDIA ConnectX-7 VPI ➤ Up to 400 Gb/s NVIDIA InfiniBand/Ethernet 2x dual-port QSFP112 NVIDIA BlueField-3 DPU ➤ Up to 400 Gb/s NVIDIA InfiniBand/Ethernet





Cerebras: *a Wafer Scale Engine (WSE)* (Aug'19)



Cerebras Wafer Scale Engine (WSE):
the largest chip ever built

46,225 mm² chip

56x larger than the biggest GPU ever made

400,000 core

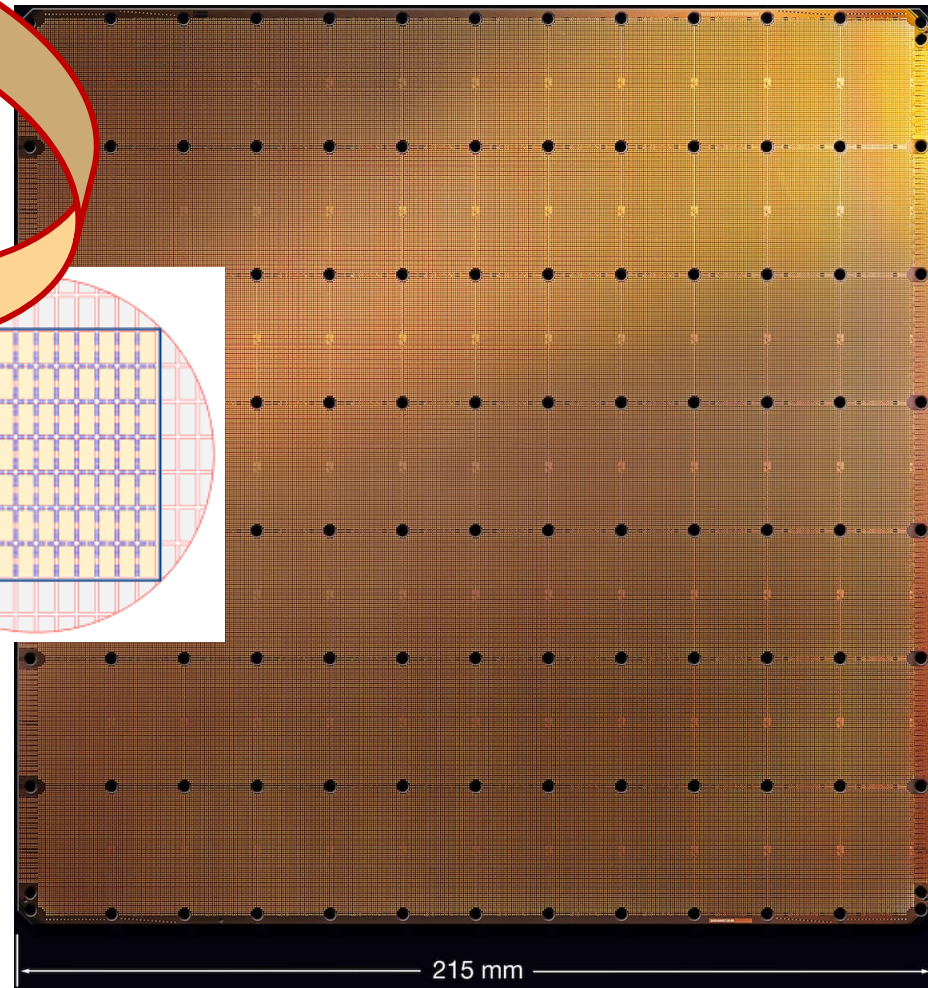
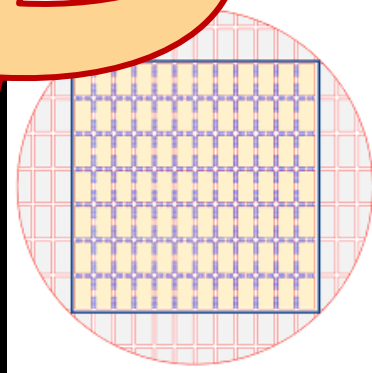
78x more cores

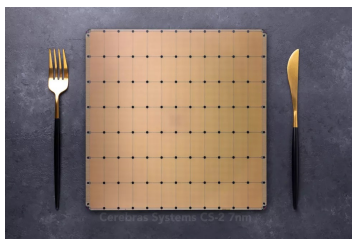
18 GB on-chip SRAM

3000x more on-chip memory

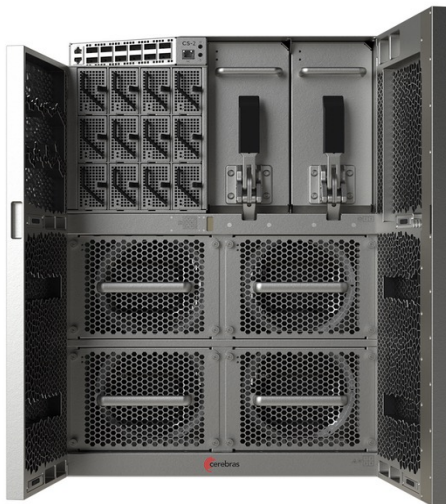
100 Pb/s interconnect

33,000x more bandwidth





Cerebras: 2nd Gen Wafer Scale Engine (WSE2) (Q3 2021)



APereira, ACA

	Cerebras Wafer Scale Engine 2	Cerebras Wafer Scale Engine	Nvidia A100
Process Node	TSMC 7nm	TSMC 16nm	TSMC 7nm N7
AI Cores	850,000	400,000	6,912 + 432
Die Size	46,255 mm ²	46,255 mm ²	826 mm ²
Transistors	2.6 Trillion	1.2 Trillion	54 Billion
On-Chip SRAM Memory	40 GB	18 GB	40 MB
Memory Bandwidth	20 PB/s	9 PB/s	1,555 GB/s
Fabric Bandwidth	220 Pb/s	100 Pb/s	600 GB/s
Power Consumption (System/Chip)	20kW / 15kW	20kW / 15kW	250W (PCIe) / 400W (SXM)